2020 AMC 10A Problem 3 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 3

Problem:

Assuming $a \neq 3, b \neq 4$, and $c \neq 5$, what is the value in simplest form of the following expression?

$$\frac{a-3}{5-c} \cdot \frac{b-4}{3-a} \cdot \frac{c-5}{4-b}$$

Answer Choices:

A. -1

B. 1

C. $\frac{abc}{60}$

D. $\frac{1}{abc} - \frac{1}{60}$

E. $\frac{1}{60} - \frac{1}{abc}$

Solution:

If $x \neq y$, then

$$\frac{x-y}{y-x} = -1$$

We use this fact to simplify the original expression:

$$\begin{aligned} & \frac{a-3}{5-c} \cdot \frac{b-4}{3-a} \cdot \frac{c-5}{4-b} \\ = & \frac{a-3}{3-a} \cdot \frac{b-4}{4-b} \cdot \frac{c-5}{5-c} \\ = & -1 \cdot -1 \cdot -1 \\ = & \boxed{(A) -1} \end{aligned}$$